

Books Data Analysis Project

1. Libraries Import

- Pandas
- Matplotlib
- Seaborn
- Purpose: Data manipulation and visualization.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

✓ 9.7s

2. Dataset Loading

- Loaded the books dataset using pandas.

Importing Data

```
df = pd.read_csv("Books_Data_Clean.csv")
```

3. Initial Data Inspection

- Viewed sample records using head().
- Generated descriptive statistics using describe().

VIEW : Top 5 Rows

```
df.head()
```

Outputs are collapsed ...

CHECK : Summary of Dataset

```
df.describe()
```

Outputs are collapsed ...

```
df.describe()
```

	index	Publishing Year	Book_average_rating	Book_ratings_count	gross sales	publisher revenue	sale price	sales rank	units sold
count	1070.000000	1069.000000	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000
mean	534.500000	1971.377923	4.007000	94909.913084	1856.622944	843.281030	4.869561	611.652336	9676.980374
std	309.026698	185.080257	0.247244	31513.242518	3936.924240	2257.596743	3.559919	369.849830	15370.571306
min	0.000000	-560.000000	2.970000	27308.000000	104.940000	0.000000	0.990000	1.000000	106.000000
25%	267.250000	1985.000000	3.850000	70398.000000	372.465000	0.000000	1.990000	287.500000	551.250000
50%	534.500000	2003.000000	4.015000	89309.000000	809.745000	273.078000	3.990000	595.500000	3924.000000
75%	801.750000	2010.000000	4.170000	113906.500000	1487.957500	721.180500	6.990000	932.500000	5312.250000
max	1069.000000	2016.000000	4.770000	206792.000000	47795.000000	28677.000000	33.860000	1273.000000	61560.000000

4. Data Cleaning

- Filtered invalid publishing years.

FIX : Publishing Year min value (-560.000000)

```
df = df[df["Publishing Year"] > 1900]
```

- Checked missing values.

```
CHECK : Missing Values

df.isna().sum()

index          0
Publishing Year 0
Book Name      21
Author         0
language_code  49
Author_Rating  0
Book_average_rating 0
Book_ratings_count 0
genre          0
gross sales    0
publisher revenue 0
sale price     0
sales rank     0
Publisher      0
units sold     0
dtype: int64
```

- Removed records with missing book names.

```
Remove : Missing Value Rows

df.dropna(subset="Book Name",inplace = True)
```

- Checked duplicate records.

```

CHECK : Duplicate Records/Rows

df.duplicated().sum()

np.int64(0)

```

- Examined unique values in each column.

```

CHECK : Unique Records

df.nunique()

[14]
... index          988
Publishing Year    101
Book Name         987
Author            669
language_code      8
Author_Rating      4
Book_average_rating 133
Book_ratings_count 983
genre             4
gross sales        774
publisher revenue  570
sale price         143
sales rank         818
Publisher          9
units sold         470
dtype: int64

```

5. Exploratory Data Analysis

- Publishing Year Distribution (Histogram)

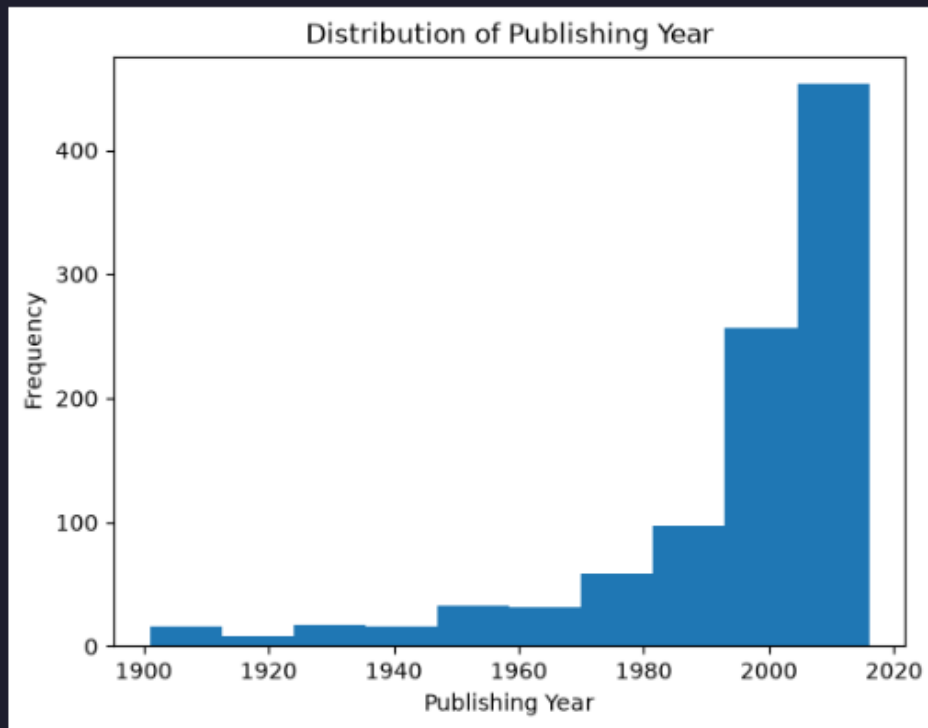
CREATE : Histogram of Publishing Year

▶

```
plt.hist(df["Publishing Year"])  
plt.xlabel("Publishing Year")  
plt.ylabel("Frequency")  
plt.title("Distribution of Publishing Year")  
plt.show()
```

[16]

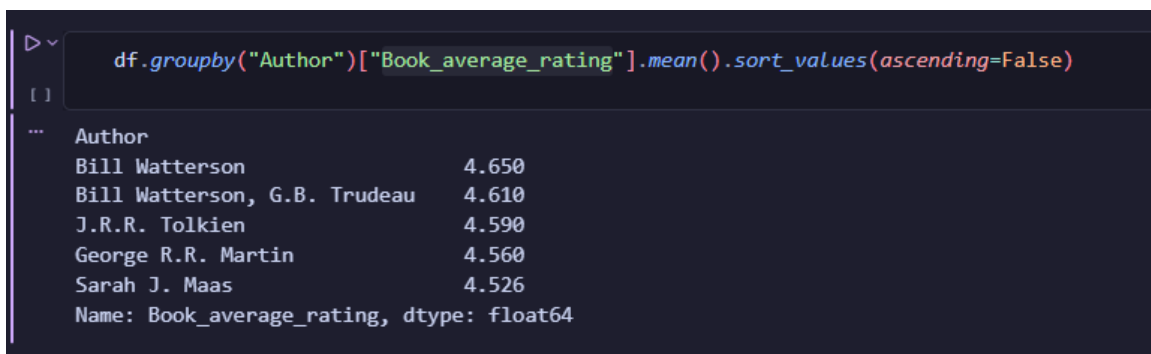
...



- Genre Distribution (Bar Chart)



- Average Rating by Author



- Book Rating Count by Genre (Box Plot)

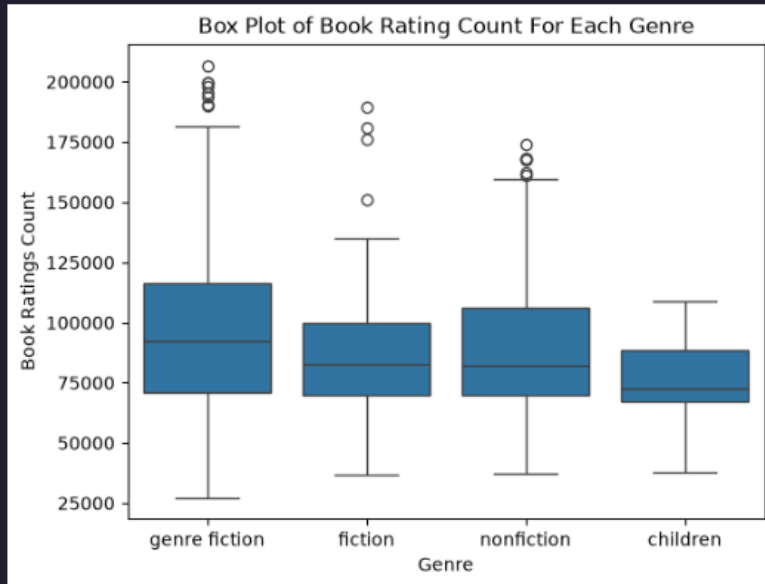
```

sns.boxplot(x= "genre" ,y="Book_ratings_count",data=df)
plt.xlabel("Genre")
plt.ylabel("Book Ratings Count")
plt.title("Box Plot of Book Rating Count For Each Genre")
plt.show()

```

[23]

...



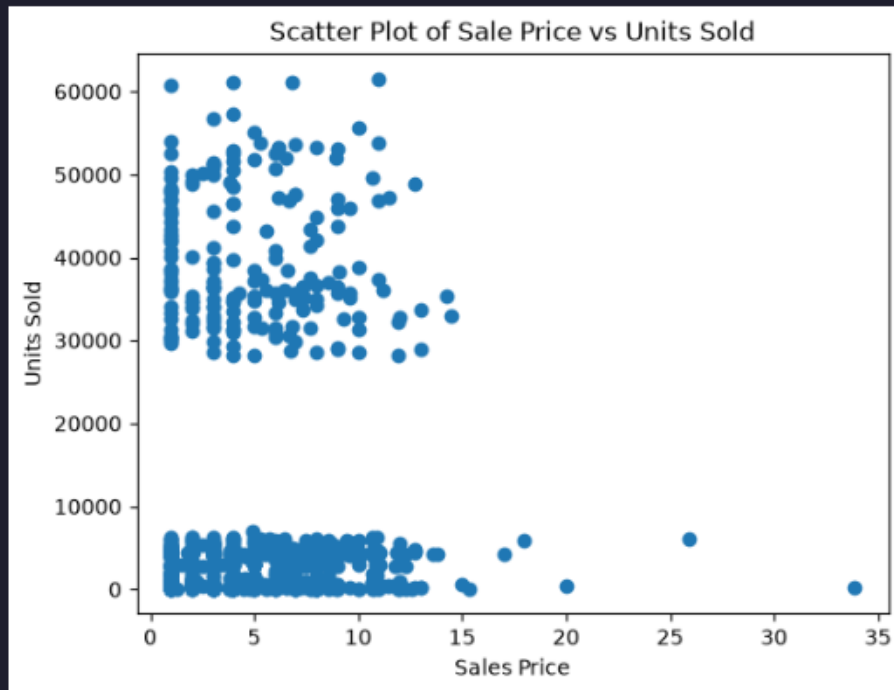
- Sale Price vs Units Sold (Scatter Plot)

▶

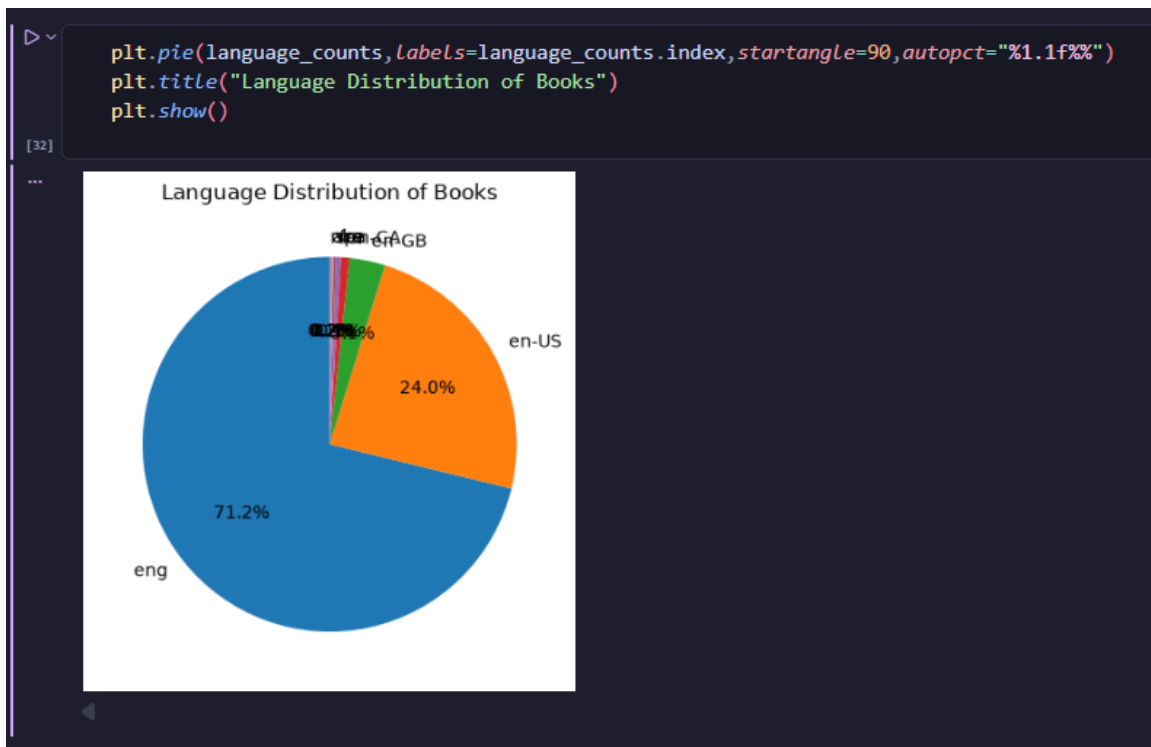
```
plt.scatter(df["sale_price"], df["units_sold"])  
plt.xlabel("Sales Price")  
plt.ylabel("Units Sold")  
plt.title("Scatter Plot of Sale Price vs Units Sold")  
plt.show()
```

[24]

...



- Language Distribution (Pie Chart)



- Publisher Revenue Analysis

```
df.groupby("Publisher ")[["publisher revenue"].sum().sort_values(ascending=False)
```

[39]

```

Publisher
Penguin Group (USA) LLC      191581.104
Random House LLC            174956.244
Amazon Digital Services, Inc. 141767.772
HarperCollins Publishers    121769.814
Hachette Book Group         107410.968
Simon and Schuster Digital Sales Inc 46858.206
Macmillan                   31249.830
HarperCollins Publishing     2830.806
HarperCollins Christian Publishing 2135.670
Name: publisher revenue, dtype: float64

```

- Author Rating Category Analysis

```
df.groupby("Author_Rating ")[["Book_ratings_count"].mean().sort_values(ascending=False)
```

[11] ✓ 0.0s

```

Author_Rating
Intermediate    101710.152921
Famous          97306.470588
Novice          87318.464286
Excellent       83529.591954
Name: Book_ratings_count, dtype: float64

```

- Language Frequency Analysis

```
[45] df.groupby("language_code").size().sort_values(ascending=False)

... language_code
eng      670
en-US    226
en-GB     29
en-CA      7
fre        4
ara         2
spa         2
nl          1
dtype: int64
```

- Variance of Ratings

```
[48] df.groupby("Author_Rating")["Book_ratings_count"].var()

... Author_Rating
Excellent      4.419857e+08
Famous         1.227555e+09
Intermediate   1.170331e+09
Novice          9.523157e+08
Name: Book_ratings_count, dtype: float64
```

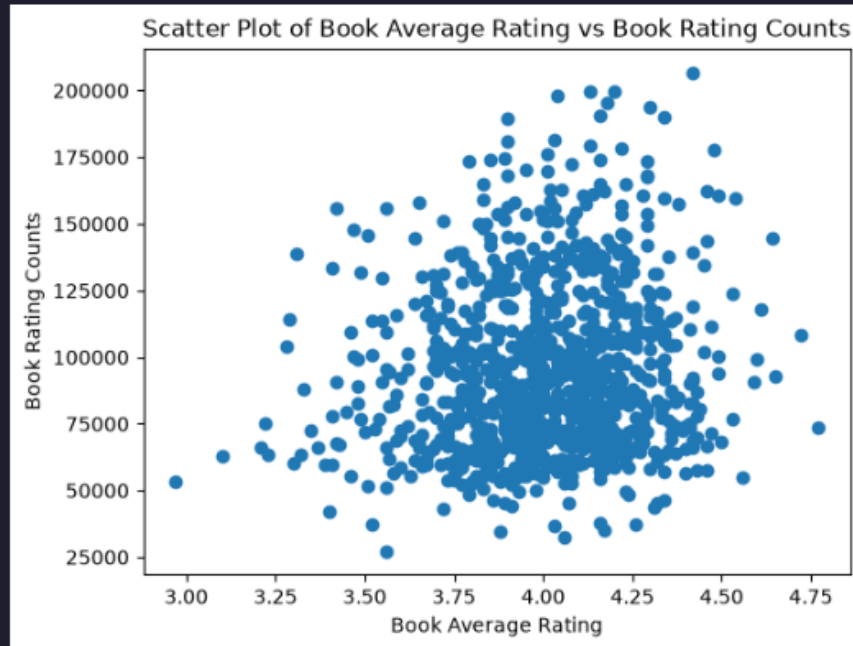
- Average Rating vs Rating Count

▷

```
plt.scatter(df["Book_average_rating"],df["Book_ratings_count"])  
plt.xlabel("Book Average Rating")  
plt.ylabel("Book Rating Counts")  
plt.title("Scatter Plot of Book Average Rating vs Book Rating Counts")  
plt.show()
```

[51]

...



- Gross Sales by Author

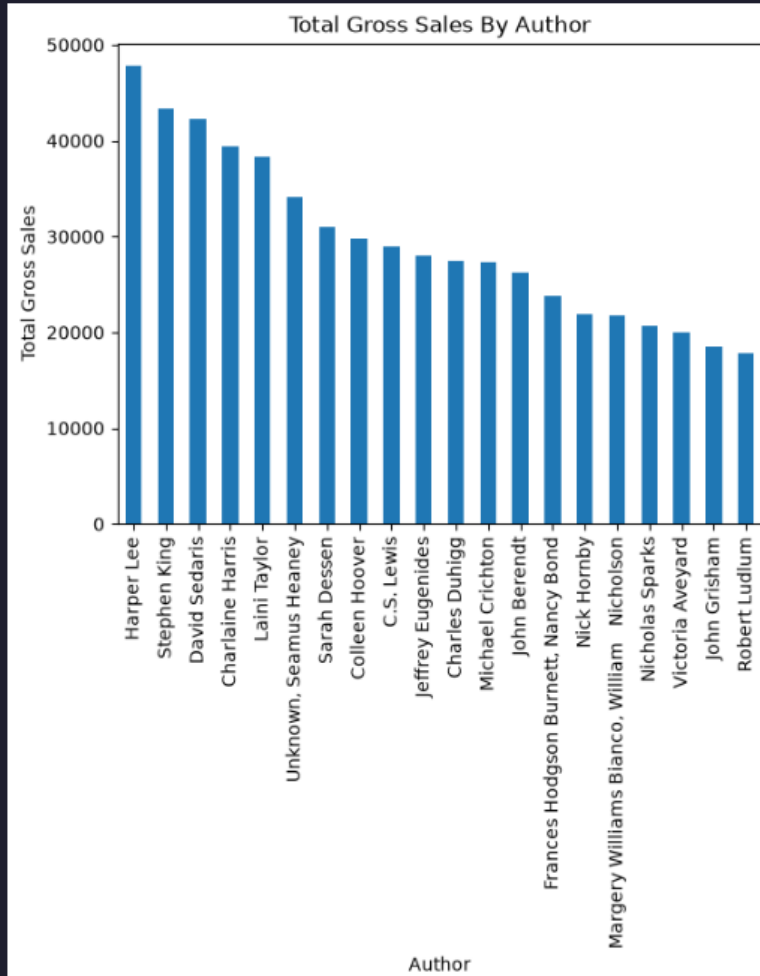
```
total_gross_sales_by_author = df.groupby("Author")["gross sales"].sum()
```

[52]

```
total_gross_sales_by_author.sort_values(ascending=False).head(20).plot(kind="bar")
plt.title("Total Gross Sales By Author")
plt.xlabel("Author")
plt.ylabel("Total Gross Sales")
plt.show()
```

[61]

...



- Units Sold Over Time



```
df.groupby("Publishing Year")["units sold"].sum().plot(kind="line")  
plt.xlabel("Publishing Year")  
plt.ylabel("Total Units Sold")  
plt.title("Total Units Sold Ove The Years")  
plt.show()
```

[64]

...

